CLAUDE IMPACT LAB 2026 · NEUROSCIENCE

Shift Happens

Do the sleep-reactivation methods built for spikes still work on slow calcium imaging? An open toolbox that runs four standard methods and only lets a method vote after it passes planted-truth tests.

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github.com/BatuCani/hackathon-2026-11-07

THE PROBLEM

Methods built for spikes, data that is slow

- Does sleep replay the co-activity patterns of the waking period before it?
- Four standard methods: cell-pair correlation, explained variance, PCA and ICA assemblies.
- All built and benchmarked on sparse spikes.
- Calcium imaging is slow and autocorrelated: their null tests may be too lenient. Nobody has checked.

Tingley & Peyrache 2020, section 3b-v, names the gaps itself:

> *"A systematic review and analysis of this possibility is still missing in the field."*

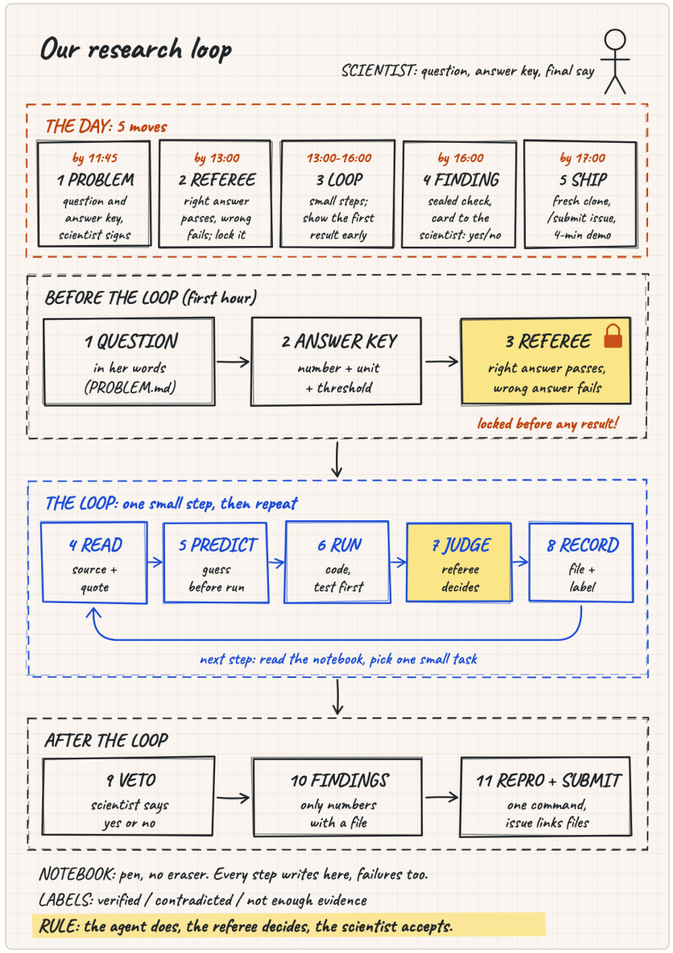
> *"a systemic variation of bin size and its impact on pattern detection is lacking."*

Calcium records thousands of identified cells across days. A valid test opens sleep-memory questions at a scale electrodes cannot reach.

THE LOOP AND THE REFEREE

Answer key first, referee second, code last

- **Locked answer key** (PROBLEM.md): one number, unit and yes-threshold per method, before any result.
- One null, N1: each neuron's trace circularly shifted by its own random offset, 500 times, seed 0.
- **Referee**: synthetic calcium with planted truth - reactivation, pure noise, assemblies already in PRE, a firing-rate confound. Fail it, no vote.
- Every step: read, predict, run, judge, record. Predictions committed before the run.



KEY FINDING, SYNTHETIC DATA

The spike-era test is too lenient for calcium

Pure calcium-like noise (tau 1 s, no assemblies), 50 seeds:

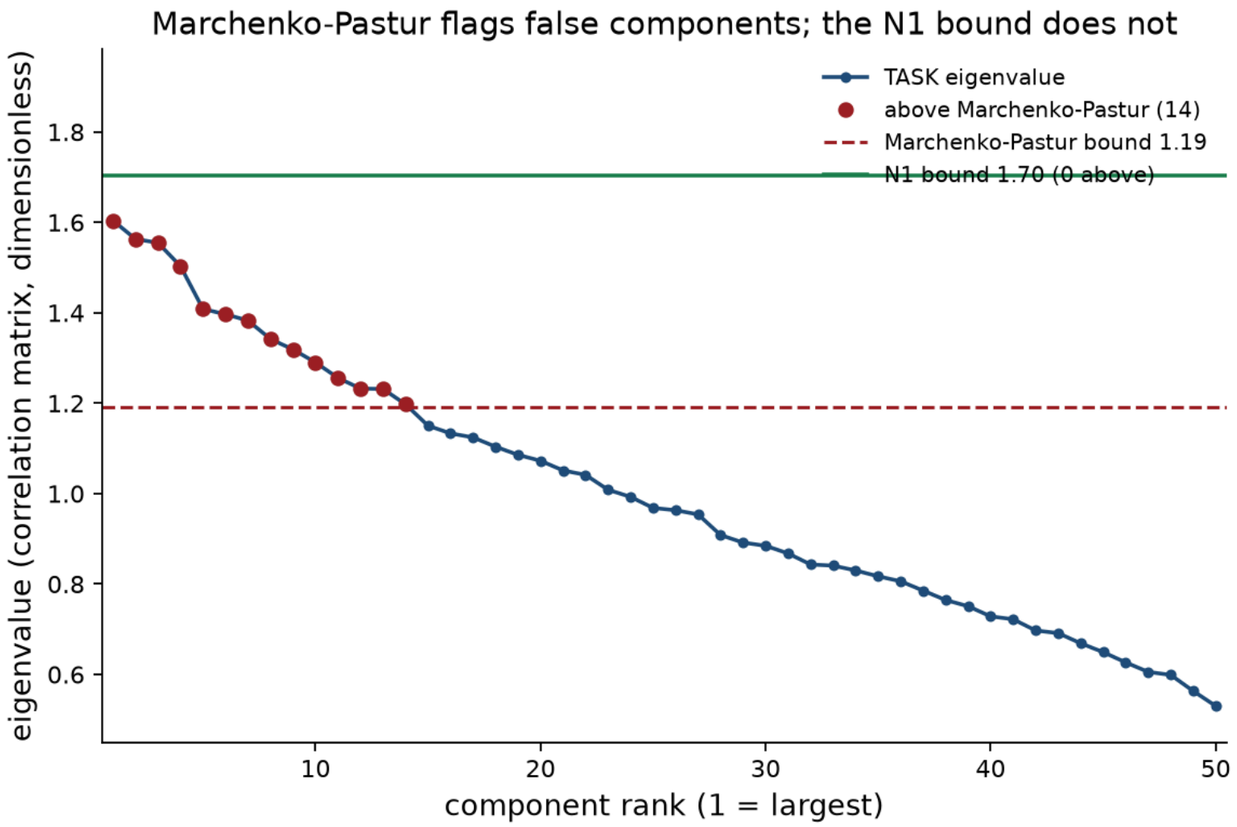
50 / 50

runs with false assemblies under the analytical Marchenko-Pastur bound (at least 5 false components)

2 / 50

runs with any false assembly under our N1 shuffle bound

Figure, seed 0: 14 eigenvalues above Marchenko-Pastur, 0 above N1. Slow calcium alone breaks the standard bound.



Synthetic TASK, no assemblies, tau_s = 1 s, N = 50 cells, B = 6000 frames, seed 0, 200 N1 shuffles (T-N1-8).

REAL DATA

Yes where it should be, no claim where it can't

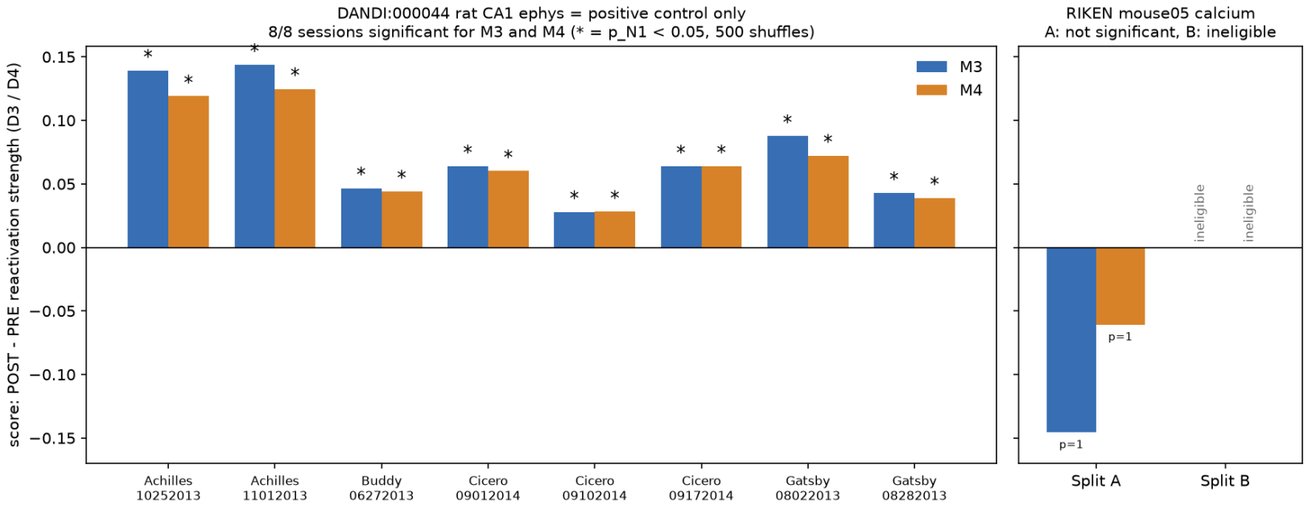
8 / 8

DANDI:000044 sessions where M3 says yes (D3 = +0.028 to +0.144, p_N1 = 0.002, the minimum). Positive control only: rat CA1 ephys.

Not enough evidence

RIKEN mouse cortex calcium (mouse05): only M1 counts in both splits, rule needs two. No counted method says yes.

Split A: D1 = -0.069, D3 = -0.146 (p_N1 = 1.0) - patterns stronger in sleep *before* the wake bout. DANDI Loop 2: M1 yes 6/8, M2 yes 7/8. Loop 3: rate confound not supported (r = 0.136, p = 0.749).



HONEST LIMITS

Where it got stuck, what it needs next

Stuck

- One public calcium session, no real task: a wake bout stands in for TASK.
- M2 fails its own control (6/10 false yes with assemblies in PRE).
- M4 fails T-M4-1 (PCA 0.817, needs < 0.8).
- Split B too short for M3/M4 (651 TASK frames < 1500).
- Every failure kept as a failure; no threshold moved.

Next

- Calcium sessions with a real learning task, several animals.
- A known-positive real calcium dataset to check the referee on calcium.
- A null that fixes M2 (block-swap at the right length); M5-M8.
- Scientist sign-off on the answer key.
- Verdict wired into run.py: one command, .mat to verdict.

Reproduce: clone, one Zenodo zip, `run.py --seed 0` (61 s, byte-identical results).